

Status	Finished
Started	Thursday, 6 November 2025, 12:00 PM
Completed	Thursday, 6 November 2025, 12:16 PM
Duration	15 mins 41 secs
Marks	2.33/3.00
Grade	2.33 out of 3.00 (77.77%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Find the solution to the initial value problem

$$\begin{cases} a_n - a_{n-1} - 6 \cdot a_{n-2} = 0 \\ a_0 = 1 \\ a_1 = -12 \end{cases}$$

$a_n =$

Input your answer as a function of n . Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **$2^n + 3$** .

Your last answer was interpreted as follows:

$$-2 \cdot 3^n + 3 \cdot (-2)^n$$

The variables found in your answer were: $[n]$

Question 2

Partially correct

Mark 0.33 out of 1.00

Consider $G = S_6$, the symmetric group of permutations of 6 symbols.

Recall that a subgroup $H \subseteq G$ is called *non-trivial* if $H \neq \{e\}$, and *proper* if $H \neq G$.

Select every statement below which is true, and leave the false statements unchecked.

- ☒ The order of G is divisible by 3.
- ☐ The group G is finite.
- ☐ The only subgroups of G are the trivial subgroup $\{e\}$, and G itself.
- ☒ There exists an element $g \in G$ such that the order of the subgroup $\langle g \rangle$ equals the order of G .
- ☒ There exist elements $g, h \in G$ such that $gh \neq hg$.
- ☐ None of the above

Question 3

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 4}$, and let $H \subset G$ be the subgroup

$$H = \{[0, 0, 0, 0], [0, 0, 0, 1], [0, 1, 1, 0], [0, 1, 1, 1], [1, 0, 1, 0], [1, 0, 1, 1], [1, 1, 0, 0], [1, 1, 0, 1]\}.$$

What is the index of $H \subset G$?

$$[G : H] = 2$$

Your last answer was interpreted as follows:

2

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that reflects around the line $(y=x)$.

Compute the index of the subgroup generated by σ :

$$[D_4 : \langle \sigma \rangle] = 4$$

Your last answer was interpreted as follows:

4

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 111

Signing code:

Your last answer was interpreted as follows:

36692

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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